Caculator in Python using Tkinter

Submitted By

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Converted from notion

Calculator Application Using Python Tkinter

1. Introduction

The Calculator Application is a Graphical User Interface (GUI) based software developed using Python and the Tkinter library. The application provides users with a simple and efficient platform for performing arithmetic calculations through an interactive and visually appealing interface.

The project demonstrates the implementation of GUI development, event-driven programming, and arithmetic expression evaluation while emphasizing usability and functionality.

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2. Problem Statement

Traditional command-line calculators lack an intuitive graphical interface, making them less convenient for everyday users. There is a need for a calculator application that combines ease of use, accessibility, and modern design while supporting essential mathematical operations.

This project aims to develop a GUI-based calculator that provides a user-friendly experience through mouse and keyboard interactions.

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3. Objectives

The primary objectives of this project are:

- Develop a graphical calculator using Python.
- Implement arithmetic operations such as addition, subtraction, multiplication, and division.

- Support keyboard-based user interaction.
- Provide percentage calculation functionality.
- Implement error handling mechanisms.
- Create a modern and visually appealing user interface.

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4. Technologies Used

Programming Language

- Python 3

Libraries

- Tkinter

Development Tools

- Visual Studio Code
- GitHub

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5. System Design

The calculator follows an event-driven architecture where user actions trigger specific functions.

Workflow

User Input

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Button Click / Keyboard Event

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Expression Processing

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Calculation Engine

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Display Result

The application captures user input through buttons or keyboard events, processes the arithmetic expression, evaluates the result, and displays the output through the GUI.

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6. Features Implemented

The calculator includes the following features:

- Addition
- Subtraction
- Multiplication
- Division
- Percentage Calculation
- Positive/Negative Toggle
- Backspace Function
- Clear Display Function
- Keyboard Support
- Error Handling
- Modern Dark-Themed Interface

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7. Implementation Details

The application was developed using Tkinter widgets such as:

- Entry Widget for displaying expressions and results.
- Button Widgets for calculator operations.
- Frame Widgets for organizing the layout.

Custom functions were implemented to:

- Handle button clicks.
- Evaluate arithmetic expressions.
- Process keyboard inputs.
- Toggle positive and negative values.
- Calculate percentages.
- Remove characters using backspace.
- Handle invalid expressions through exception handling.

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8. Testing

The application was tested using multiple arithmetic expressions to verify correctness.

Test Cases

Input Expression	Expected Output	Result
5 + 5	10	Passed
10 - 3	7	Passed
8 * 4	32	Passed
20 / 5	4	Passed
50 %	0.5	Passed
Invalid Input	Error	Passed

The calculator successfully handled all tested scenarios.

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9. Results

The developed calculator successfully performs arithmetic calculations while providing a responsive graphical user interface.

The application supports both mouse and keyboard interactions, ensuring ease of use and improved accessibility.

The modern dark-themed design enhances the overall user experience and gives the application a professional appearance.

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10. Learning Outcomes

Through this project, the following concepts were learned and reinforced:

- Python GUI Development
- Tkinter Widget Management
- Event-Driven Programming
- Function Design
- Error Handling Techniques
- User Interface Design Principles
- Software Testing and Debugging

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11. Conclusion

The Calculator Application was successfully developed using Python and Tkinter. The project demonstrates the practical implementation of GUI programming concepts while providing a functional and user-friendly calculator.

The application meets all project objectives by supporting arithmetic calculations, keyboard interaction, percentage calculations, error handling, and modern interface design. The project serves as a strong example of desktop application development using Python.